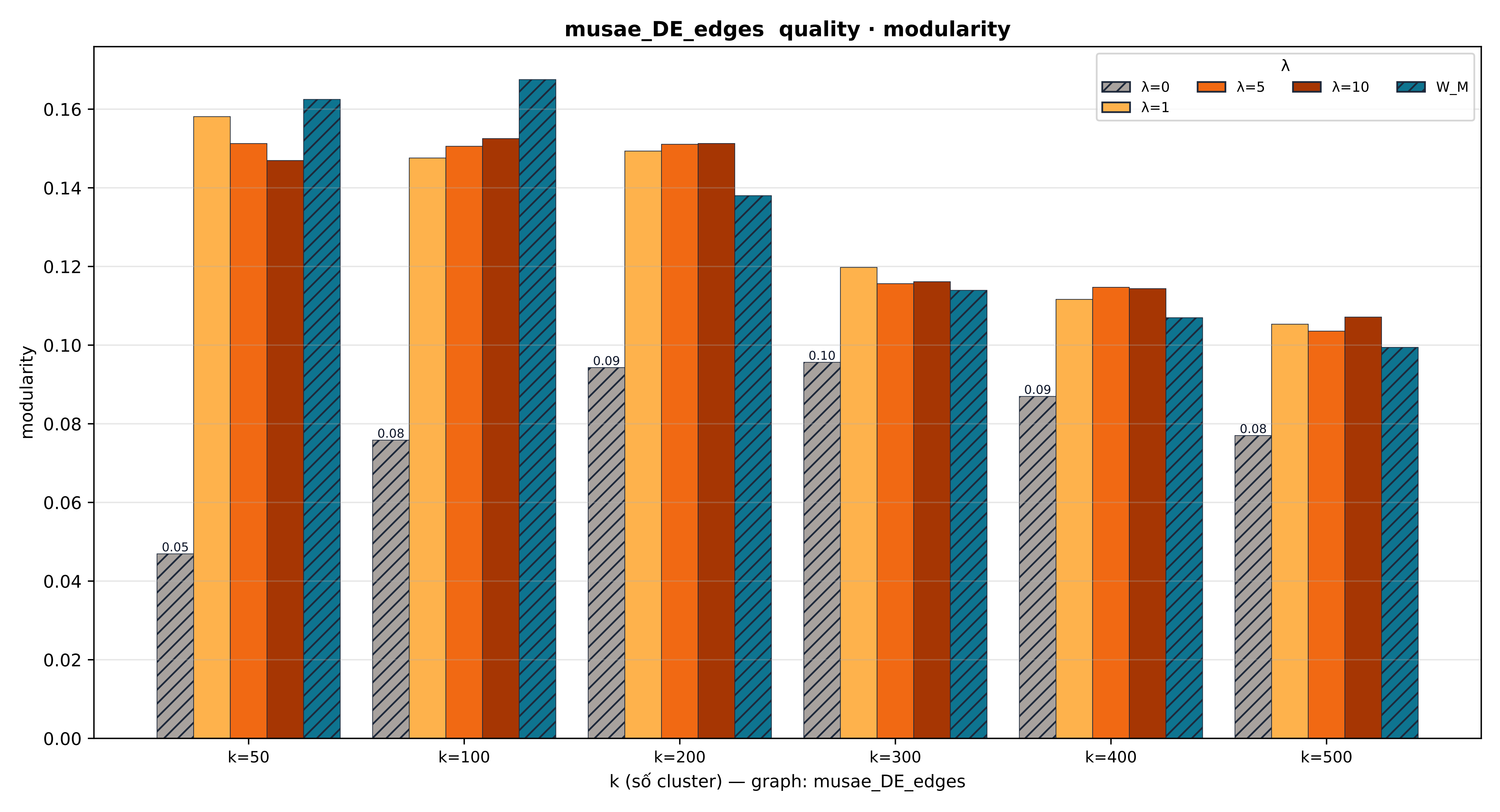
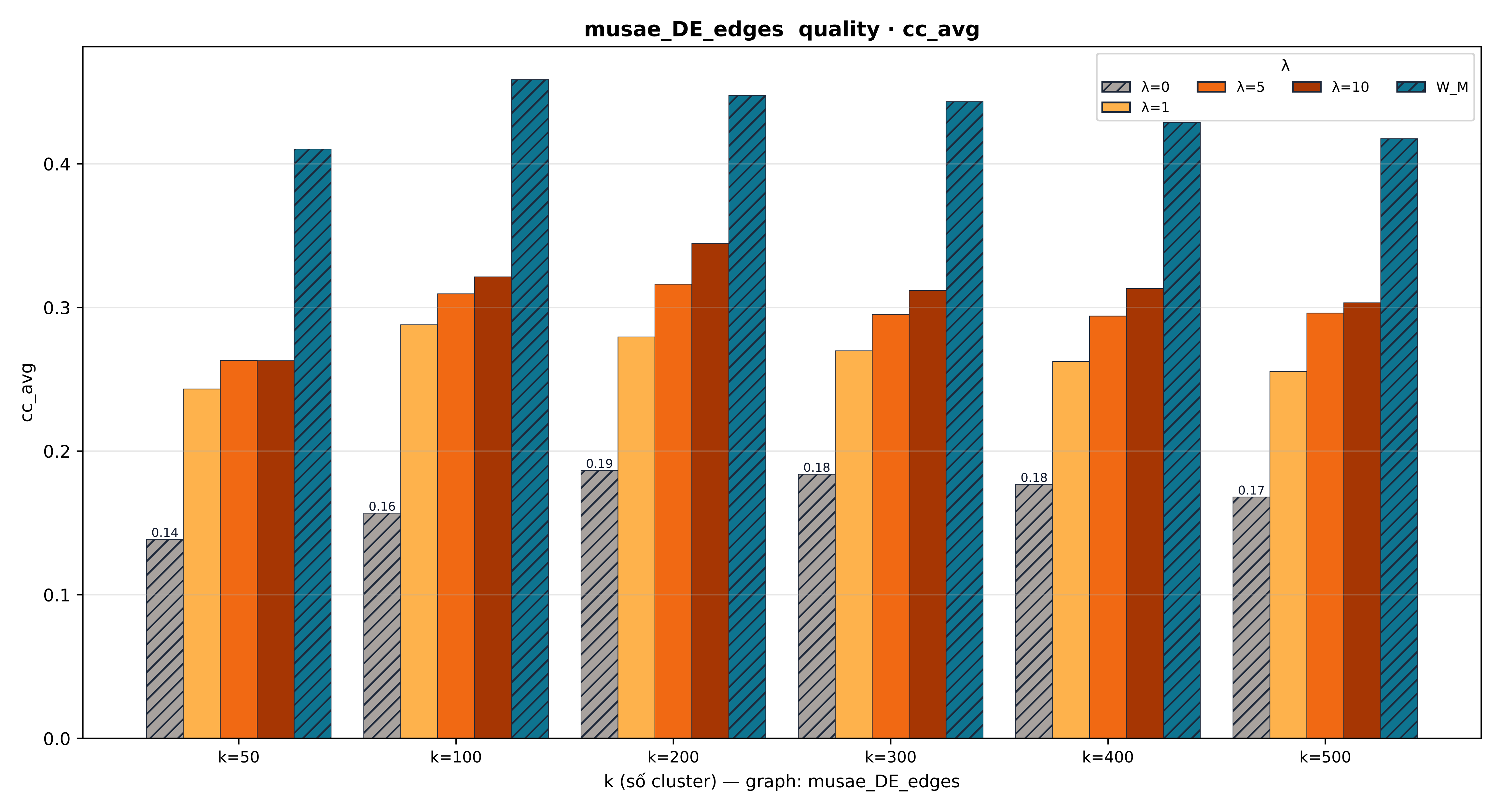
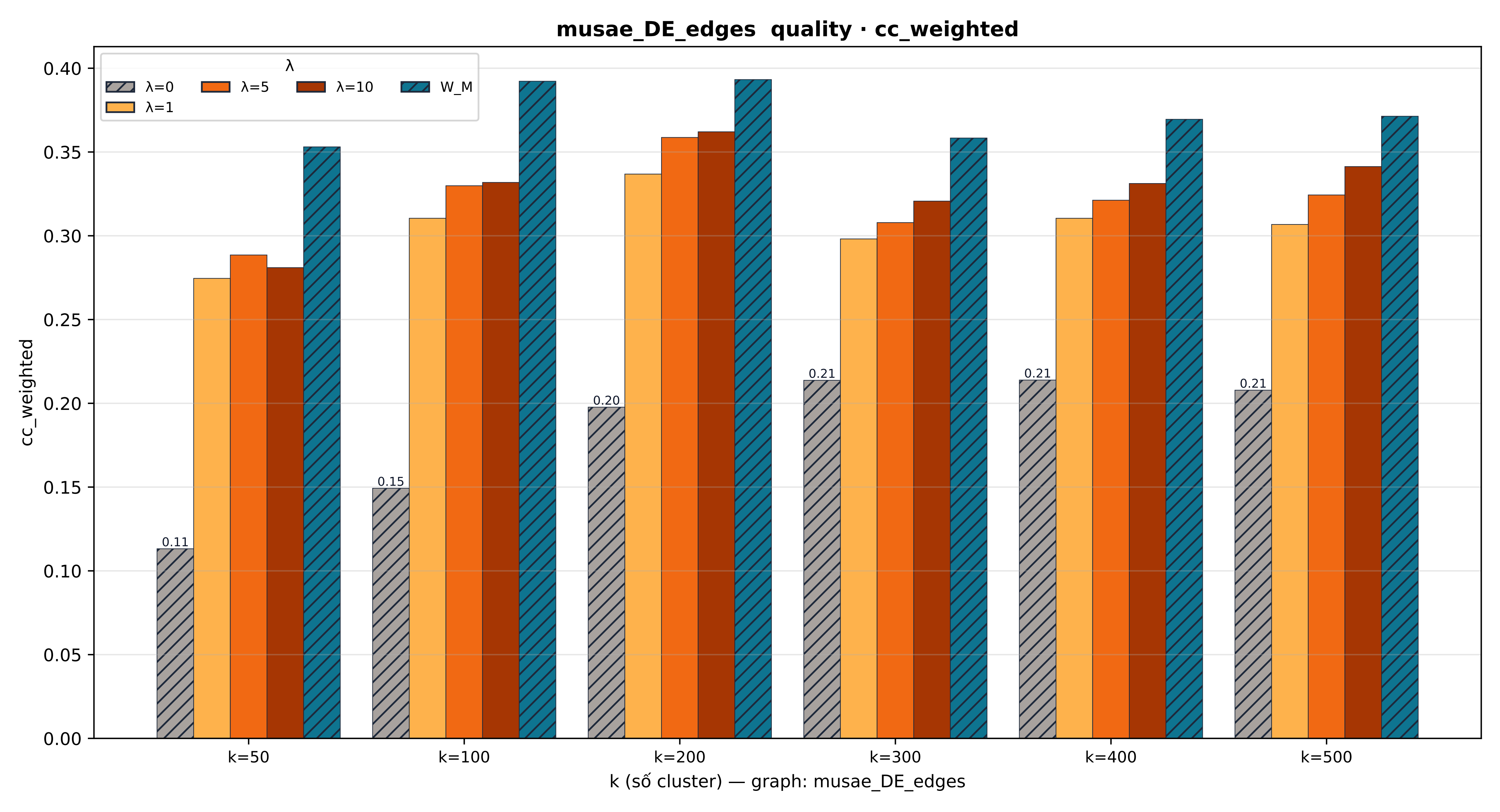
MAIN RESULT – REAL
Real-world dataset (SNAP / MUSAE)

Top 6 bộ test có Q cải thiện mạnh nhất khi dùng mixed matrix W_λ
(filter: Q(A) ≥ 0.1, score = max Q(λ>0) − Q(A))

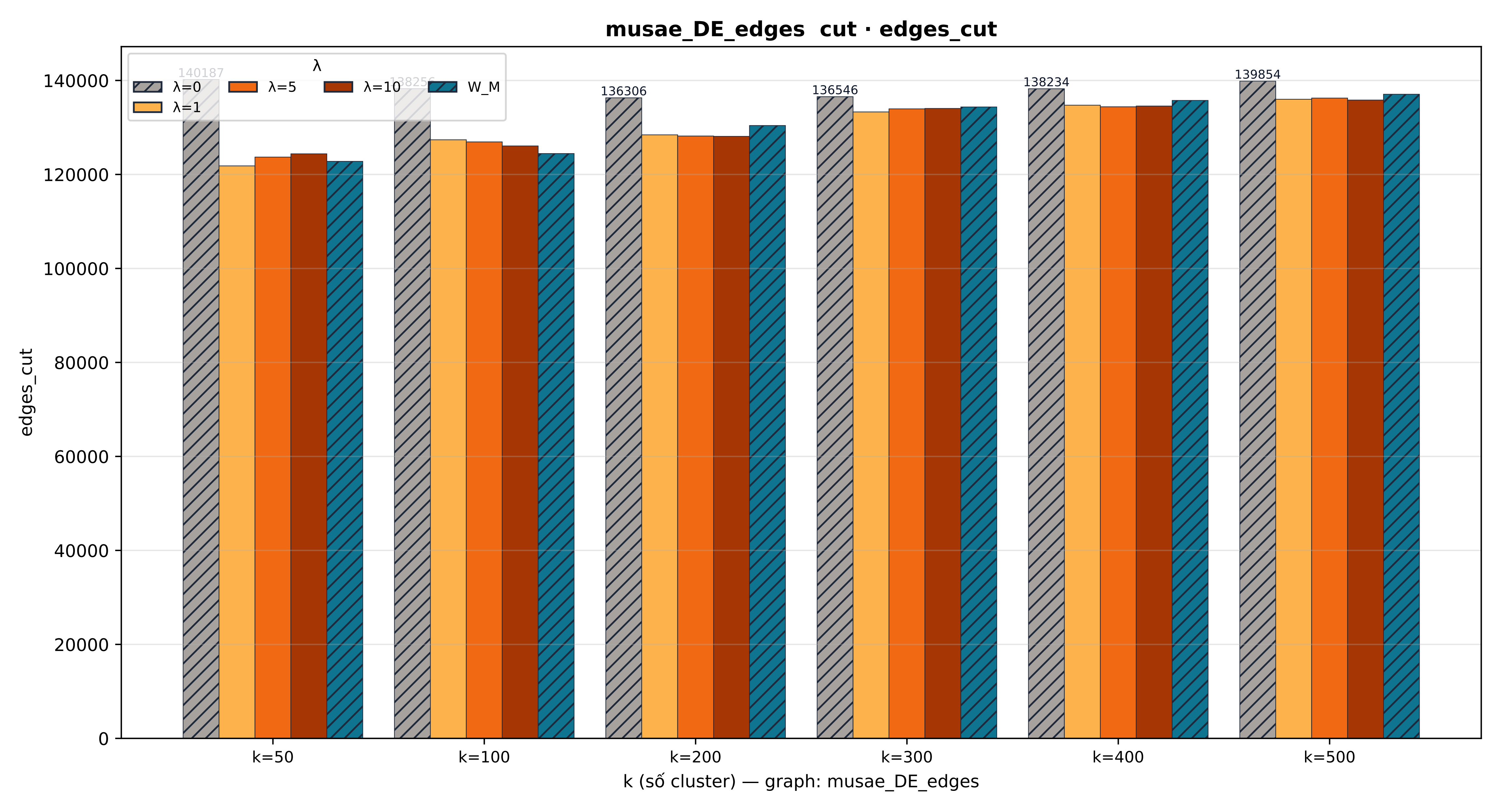
```
1. musae_DE_edges-ncut-k50              Q(A)=0.0469  →  Q(λ=W_M)=0.1625  ΔQ=+0.1156
2. musae_DE_edges-ncut-k100             Q(A)=0.0758  →  Q(λ=W_M)=0.1675  ΔQ=+0.0917
3. musae_DE_edges-ncut-k200             Q(A)=0.0943  →  Q(λ=10.0)=0.1513  ΔQ=+0.0570
4. musae_DE_edges-ncut-k300             Q(A)=0.0956  →  Q(λ=1.0)=0.1198  ΔQ=+0.0241
5. musae_DE_edges-ncut-k400             Q(A)=0.0870  →  Q(λ=5.0)=0.1147  ΔQ=+0.0277
6. musae_DE_edges-ncut-k500             Q(A)=0.0770  →  Q(λ=10.0)=0.1071  ΔQ=+0.0301
```

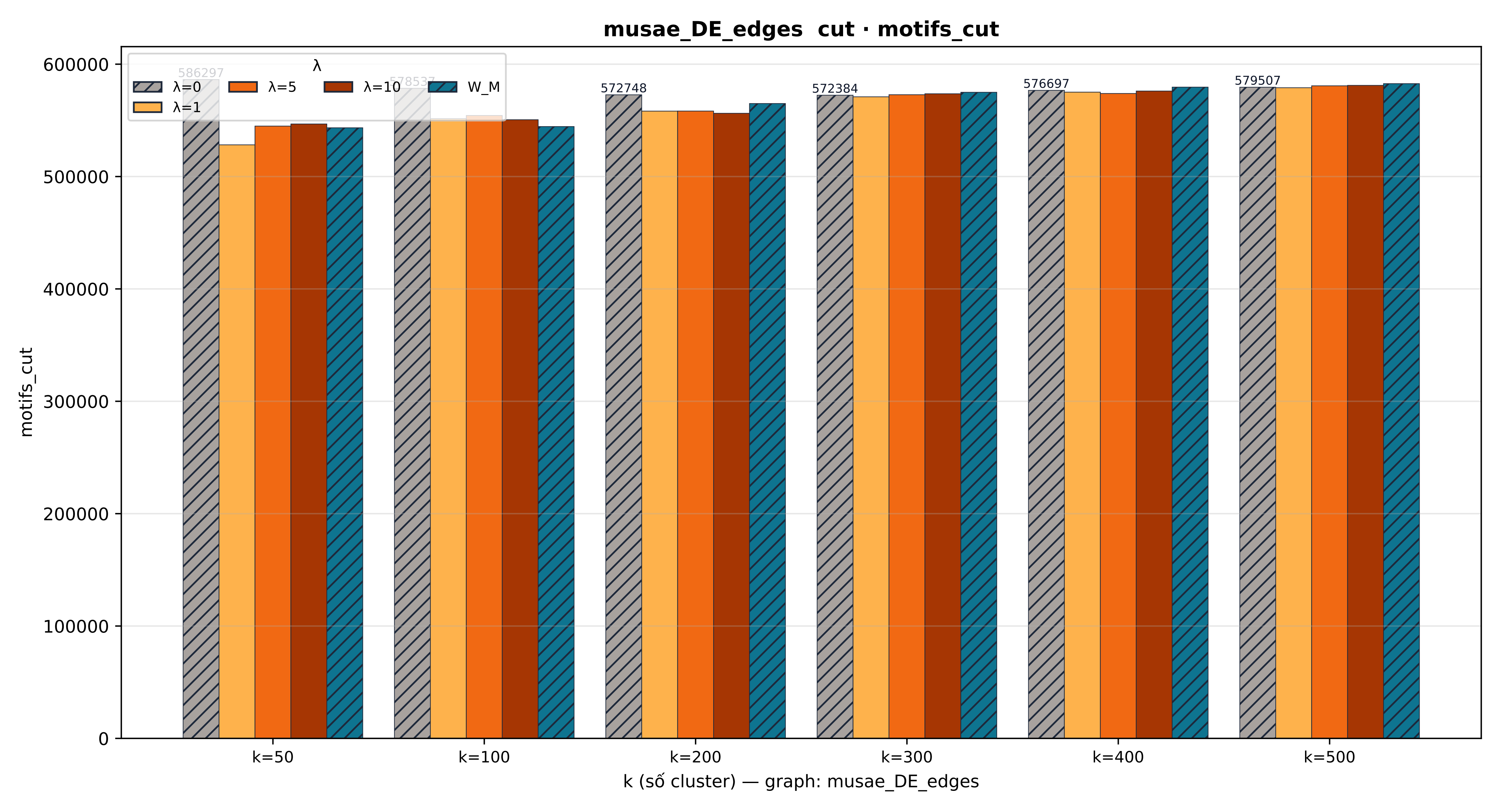

musae_DE_edges quality · modularity
λ
λ=0
λ=1
λ=5
λ=10
W_M
modularity
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
0.05
0.08
0.09
0.10
0.09
0.08
k=50
k=100
k=200
k=300
k=400
k=500
k (số cluster) — graph: musae_DE_edges


musae_DE_edges quality · cc_avg
λ
λ=0
λ=1
λ=5
λ=10
W_M
cc_avg
0.0
0.1
0.2
0.3
0.4
0.14
0.16
0.19
0.18
0.18
0.17
k=50
k=100
k=200
k=300
k=400
k=500
k (số cluster) — graph: musae_DE_edges

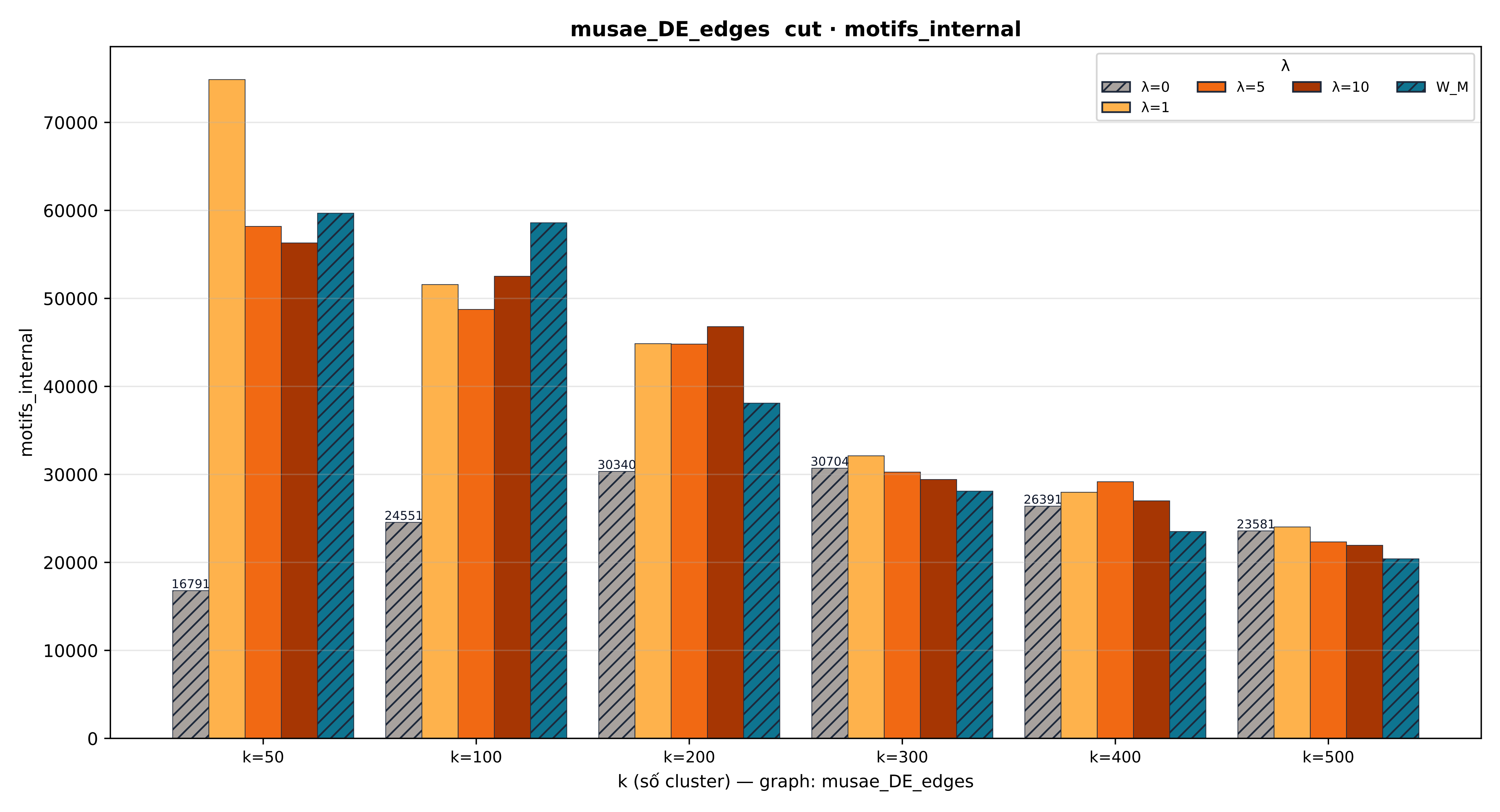


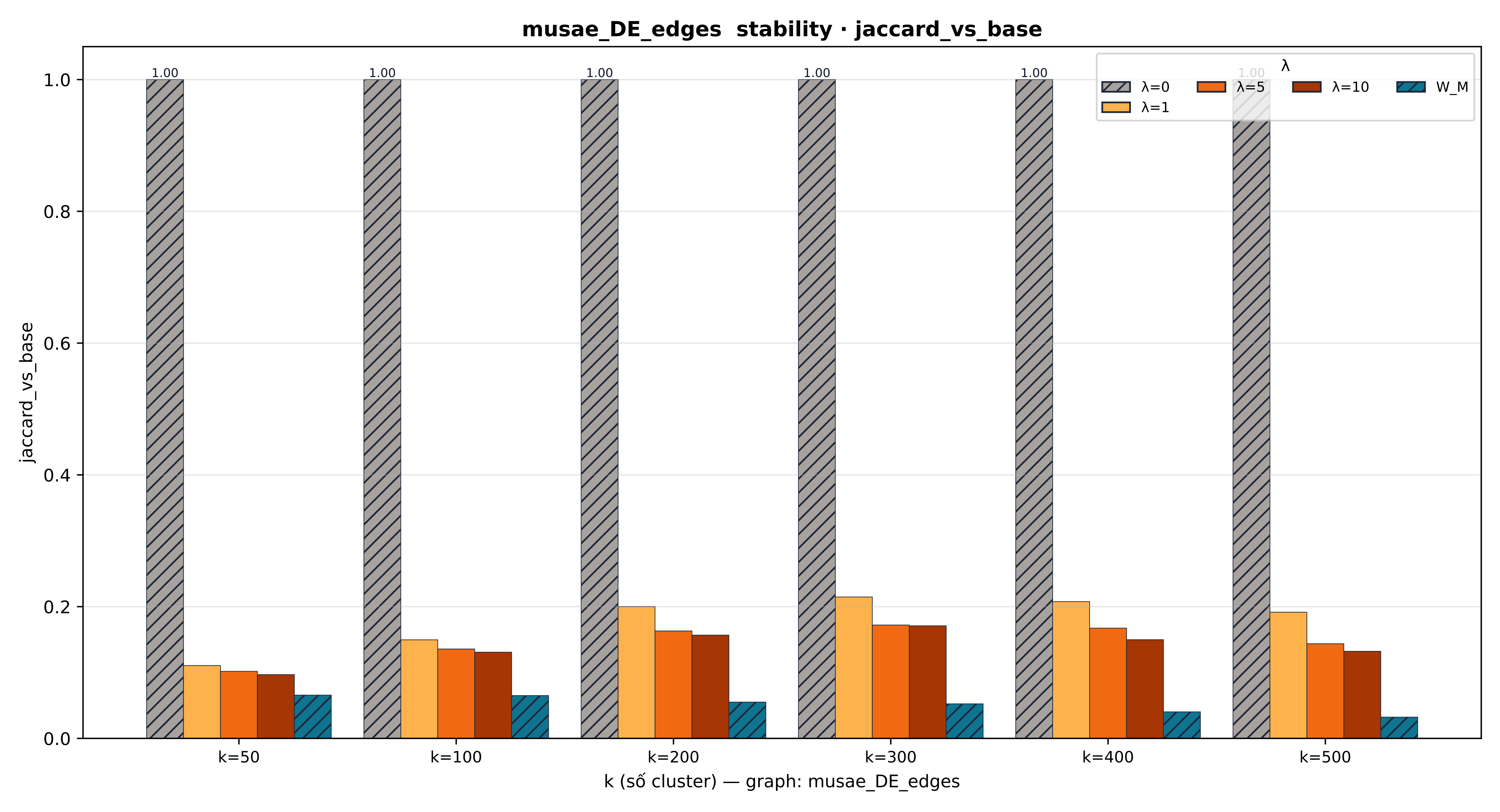
musae_DE_edges quality · cc_weighted
λ
λ=0
λ=1
λ=5
λ=10
W_M
cc_weighted
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.11
0.15
0.20
0.21
0.21
0.21
k=50
k=100
k=200
k=300
k=400
k=500
k (số cluster) — graph: musae_DE_edges


musae_DE_edges cut · edges_cut
λ
λ=0
λ=1
λ=5
λ=10
W_M
140187
138256
136306
136546
138234
139854
edges_cut
0
20000
40000
60000
80000
100000
120000
140000
k=50
k=100
k=200
k=300
k=400
k=500
k (số cluster) — graph: musae_DE_edges



musae_DE_edges cut · motifs_cut
λ
λ=0
λ=1
λ=5
λ=10
W_M
586297
578537
572748
572384
576697
579507
600000
500000
400000
300000
200000
100000
0
motifs_cut
k=50
k=100
k=200
k=300
k=400
k=500
k (số cluster) — graph: musae_DE_edges


musae_DE_edges cut · motifs_internal
λ
λ=0
λ=5
λ=10
W_M
λ=1
motifs_internal
0
10000
20000
30000
40000
50000
60000
70000
16791
24551
30340
30704
26391
23581
k=50
k=100
k=200
k=300
k=400
k=500
k (số cluster) — graph: musae_DE_edges


musae_DE_edges stability · jaccard_vs_base
jaccard_vs_base
k (số cluster) — graph: musae_DE_edges
k=50
k=100
k=200
k=300
k=400
k=500
1.00
λ
λ=0
λ=1
λ=5
λ=10
W_M
0.0
0.2
0.4
0.6
0.8
1.0